

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

P R O J E C T R E P O R T

Submitted by

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JAVA ASSIGNMENT

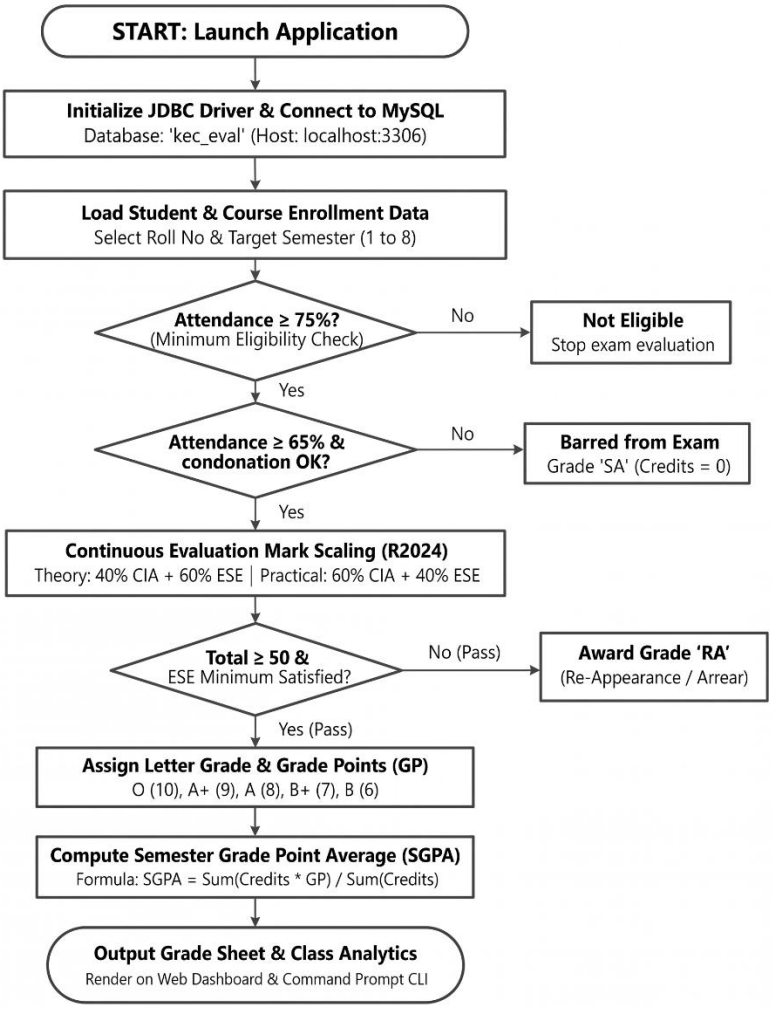
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC and MariaDB.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

FLOW CHART:



PROCEDURE:

- Load the MariaDB JDBC driver to enable communication between the Java application and the MariaDB database.

- Establish a connection to the kec_academic database running on the local MariaDB server using DriverManager.getConnection().

- Store and retrieve student details, including roll number, name, department, and semester, along with course information such as course code, title, credits, and course type.

- Enroll students in semester courses, record the number of classes held and attended, and calculate the attendance percentage.

- Verify examination eligibility based on attendance percentage. Students with 75% or above are eligible, those with 65%–74% require medical condonation verification, and those below 65% are marked as Shortage of Attendance (SA).

- Record CIA-1, CIA-2, and assignment marks, and calculate the scaled internal assessment marks according to the specified theory and practical course weightages.

- Record end-semester examination marks out of 100, verify the minimum passing requirements, and scale the marks according to the applicable theory and practical .

- Calculate the total marks by combining scaled CIA and ESE marks. Assign letter grades and grade points according to the specified grading scale, marking students as RA if they fail the examination or score below the required passing mark.

- Generate student grade sheets, attendance shortage reports, and department-wise pass percentage reports. Display the results through the interactive web dashboard and the command prompt console**.**

DATABASE SCHEMA:

1.STUDENTS TABLE AND COURSE TABLE:

```
MariaDB [kec_academic]> SELECT * FROM students;
+----------+--------------+------------+------------------+
| roll_no  | student_name | department | current_semester |
+----------+--------------+------------+------------------+
| 25ITR182 | Yuvashree    | IT         |                3 |
| 25ITR183 | Arun Kumar   | IT         |                3 |
| 25ITR184 | Priya S      | CSE        |                3 |
| 25ITR185 | Karthik R    | ECE        |                3 |
| 25ITR186 | Divya M      | IT         |                3 |
| 25ITR187 | Rahul S      | CSE        |                3 |
| 25ITR188 | Nithya R     | ECE        |                3 |
| 25ITR189 | Vignesh K    | IT         |                3 |
| 25ITR190 | Keerthana P  | CSE        |                3 |
| 25ITR191 | Santhosh B   | ECE        |                3 |
| 25ITR192 | Harini V     | IT         |                3 |
+----------+--------------+------------+------------------+
11 rows in set (0.001 sec)

MariaDB [kec_academic]> SELECT * FROM courses;
+-------------+-----------------+---------+-------------+
| course_code | course_title    | credits | course_type |
+-------------+-----------------+---------+-------------+
| 24ITR31     | Java Programming |     4.0 | THEORY      |
+-------------+-----------------+---------+-------------+
1 row in set (0.001 sec)
```

2. ENROLLMENTS TABLE:

```
+-------------+-----------------+---------+-------------+
1 row in set (0.001 sec)

MariaDB [kec_academic]> SELECT * FROM enrollments;
+---------------+----------+-------------+----------+--------------+-----------------+--------
-------------+
| enrollment_id | roll_no  | course_code | semester | total_classes | classes_attended | medical
_condonation |
+---------------+----------+-------------+----------+--------------+-----------------+--------
-------------+
|             1 | 25ITR182 | 24ITR31     |        3 |           50 |              42 |
           0 |
|             2 | 25ITR183 | 24ITR31     |        3 |           50 |              42 |
           0 |
|             3 | 25ITR184 | 24ITR31     |        3 |           50 |              45 |
           0 |
|             4 | 25ITR185 | 24ITR31     |        3 |           50 |              40 |
           0 |
|             5 | 25ITR186 | 24ITR31     |        3 |           50 |              44 |
           0 |
|             6 | 25ITR187 | 24ITR31     |        3 |           50 |              38 |
           0 |
|             7 | 25ITR188 | 24ITR31     |        3 |           50 |              47 |
           0 |
|             8 | 25ITR189 | 24ITR31     |        3 |           50 |              41 |
           0 |
|             9 | 25ITR190 | 24ITR31     |        3 |           50 |              46 |
           0 |
|            10 | 25ITR191 | 24ITR31     |        3 |           50 |              39 |
           0 |
|            11 | 25ITR192 | 24ITR31     |        3 |           50 |              43 |
           0 |
+---------------+----------+-------------+----------+--------------+-----------------+--------
-------------+
11 rows in set (0.001 sec)
```

3. ASSESSMENT MARKS TABLE:

```
MariaDB [kec_academic]> SELECT * FROM assessment_marks;
+---------------+-------+-------+---------------------+-----------+------------+
| enrollment_id | cia1  | cia2  | assignment_quiz_lab | ese_marks | ese_absent |
+---------------+-------+-------+---------------------+-----------+------------+
|             1 | 25.00 | 26.00 |               46.00 |     88.00 |          0 |
|             2 | 75.00 | 80.00 |               85.00 |     78.00 |          0 |
|             3 | 82.00 | 85.00 |               85.00 |     78.00 |          0 |
|             4 | 70.00 | 76.00 |               85.00 |     78.00 |          0 |
|             5 | 88.00 | 90.00 |               85.00 |     78.00 |          0 |
|             6 | 65.00 | 70.00 |               85.00 |     78.00 |          0 |
|             7 | 91.00 | 94.00 |               85.00 |     78.00 |          0 |
|             8 | 78.00 | 82.00 |               85.00 |     78.00 |          0 |
|             9 | 85.00 | 88.00 |               85.00 |     78.00 |          0 |
|            10 | 72.00 | 75.00 |               85.00 |     78.00 |          0 |
|            11 | 90.00 | 92.00 |               85.00 |     78.00 |          0 |
+---------------+-------+-------+---------------------+-----------+------------+
11 rows in set (0.001 sec)
```

SOURCE CODE:

MAINAPP.JAVA:

```
import java.util.*;
import java.text.SimpleDateFormat;
/**
* KEC (Autonomous) - Regulation 2024
* Student Academic & Continuous Evaluation System
* Short documentation version of MainApp.java
*/
public class MainApp {
private static final AcademicDAO dao = new AcademicDAO();
private static final Scanner sc = new Scanner(System.in);
public static void main(String[] args) {
if (!DBConnection.testConnection()) {
System.out.println("Database connection failed. Check DBConnection.java.");
return;
}
System.out.println("KEC STUDENT EVALUATION SYSTEM - R2024");
boolean running = true;
while (running) {
System.out.println("\\n1.Student 2.Course 3.Enrollment/Attendance");
System.out.println("4.Enter Marks 5.Mark Sheet 6.Defaulters");
System.out.println("7.Department Summary 0.Exit");
System.out.print("Choice: ");
switch (sc.nextLine().trim()) {
case "1": registerStudent(); break;
case "2": addCourse(); break;
case "3": enrollStudent(); break;
```

```
case "4": enterMarks(); break;
case "5":
System.out.print("Roll number: ");
String roll = sc.nextLine().trim();
System.out.print("Semester: ");
printStudentMarkSheet(roll, Integer.parseInt(sc.nextLine().trim()));
break;
case "6":
System.out.print("Semester: ");
printAttendanceDefaulters(Integer.parseInt(sc.nextLine().trim()));
break;
case "7":
System.out.print("Semester: ");
printDepartmentSummary(Integer.parseInt(sc.nextLine().trim()));
break;
case "0": running = false; break;
default: System.out.println("Invalid choice.");
}
}
System.out.println("System closed.");
}
private static void enterMarks() {
System.out.print("Roll number: "); String roll = sc.nextLine().trim();
System.out.print("Course code: "); String course = sc.nextLine().trim();
System.out.print("Semester: "); int sem = Integer.parseInt(sc.nextLine().trim());
int id = dao.getEnrollmentId(roll, course, sem);
if (id <= 0) { System.out.println("Enrollment not found."); return; }
System.out.println("1.CIA marks 2.Assignment marks 3.ESE marks");
```

```
String phase = sc.nextLine().trim();
if (phase.equals("1")) {
System.out.print("CIA-1: "); double c1 = Double.parseDouble(sc.nextLine().trim());
System.out.print("CIA-2: "); double c2 = Double.parseDouble(sc.nextLine().trim());
System.out.println(dao.recordPhase1Marks(id, c1, c2) ? "Saved." : "Failed.");
} else if (phase.equals("2")) {
System.out.print("Assignment: "); double a = Double.parseDouble(sc.nextLine().trim());
System.out.println(dao.recordPhase2Marks(id, a) ? "Saved." : "Failed.");
} else if (phase.equals("3")) {
System.out.print("Absent in ESE? (y/n): "); boolean absent =
sc.nextLine().trim().equalsIgnoreCase("y");
Double marks = null;
if (!absent) {
System.out.print("ESE marks: "); marks = Double.parseDouble(sc.nextLine().trim());
}
System.out.println(dao.recordPhase3Marks(id, marks, absent) ? "Saved." : "Failed.");
} else System.out.println("Invalid phase.");
}
```

DB CONNECTION.JAVA:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

/**
 * DBConnection - Database Connection Management for KEC Evaluation System
 * Kongu Engineering College (Autonomous) - Regulation 2024
```

```
 */
public class DBConnection {
    private static final String DEFAULT_HOST = "localhost";
    private static final String DEFAULT_PORT = "3306";
    private static final String DEFAULT_DB = "kec_academic";
    private static final String DEFAULT_USER = "root";
    private static final String DEFAULT_PASS = "0000";

    public static String getUrl() {
        String host = System.getenv("DB_HOST") != null ? System.getenv("DB_HOST") : DEFAULT_HOST;
        String port = System.getenv("DB_PORT") != null ? System.getenv("DB_PORT") : DEFAULT_PORT;
        String db = System.getenv("DB_NAME") != null ? System.getenv("DB_NAME") : DEFAULT_DB;
                    return "jdbc:mysql://" + host + ":" + port + "/" + db +
"?useSSL=false&allowPublicKeyRetrieval=true&serverTimezone=UTC";
    }

    public static String getUser() {
        return System.getenv("DB_USER") != null ? System.getenv("DB_USER") : DEFAULT_USER;
    }

    public static String getPassword() {
            return System.getenv("DB_PASSWORD") != null ? System.getenv("DB_PASSWORD") :
DEFAULT_PASS;
    }

    static {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
    } catch (ClassNotFoundException e) {
        System.err.println("[ERROR] MariaDB JDBC Driver not found in classpath: " + e.getMessage());
    }
}

/**
 * Obtains a new database connection
 */
public static Connection getConnection() throws SQLException {
    return DriverManager.getConnection(getUrl(), getUser(), getPassword());
}

/**
 * Tests the database connectivity
 */
public static boolean testConnection() {
    try (Connection conn = getConnection()) {
        return conn != null && !conn.isClosed();
    }
```

OUTPUT SCREENSHOTS:

OUTPUT 1 – MAIN MENU AND STUDENT VIEW



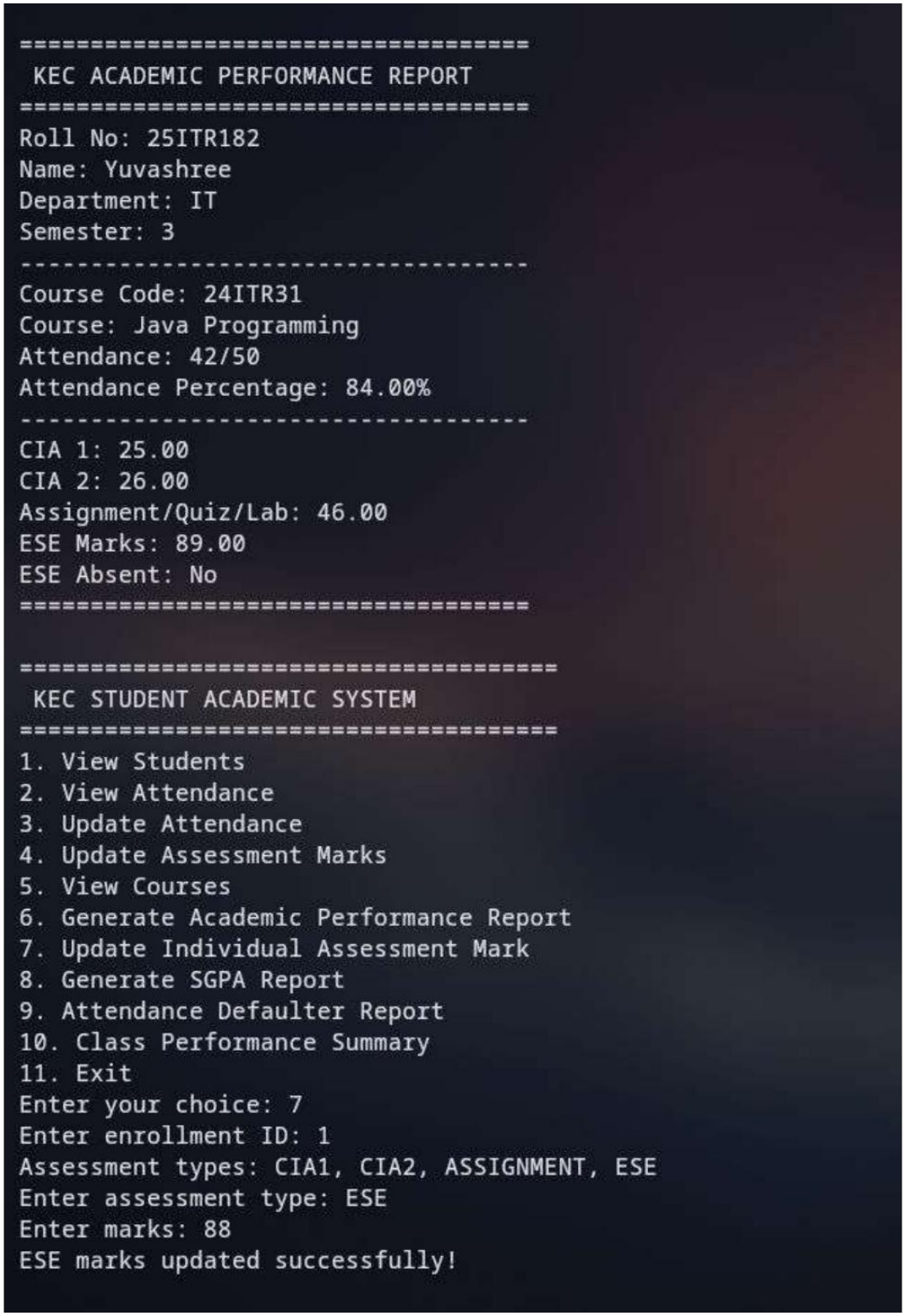
OUTPUT 2 – ATTENDANCE AND ASSESSMENT MARKS



OUTPUT 3 – COURSE AND ATTENDANCE OPERATIONS



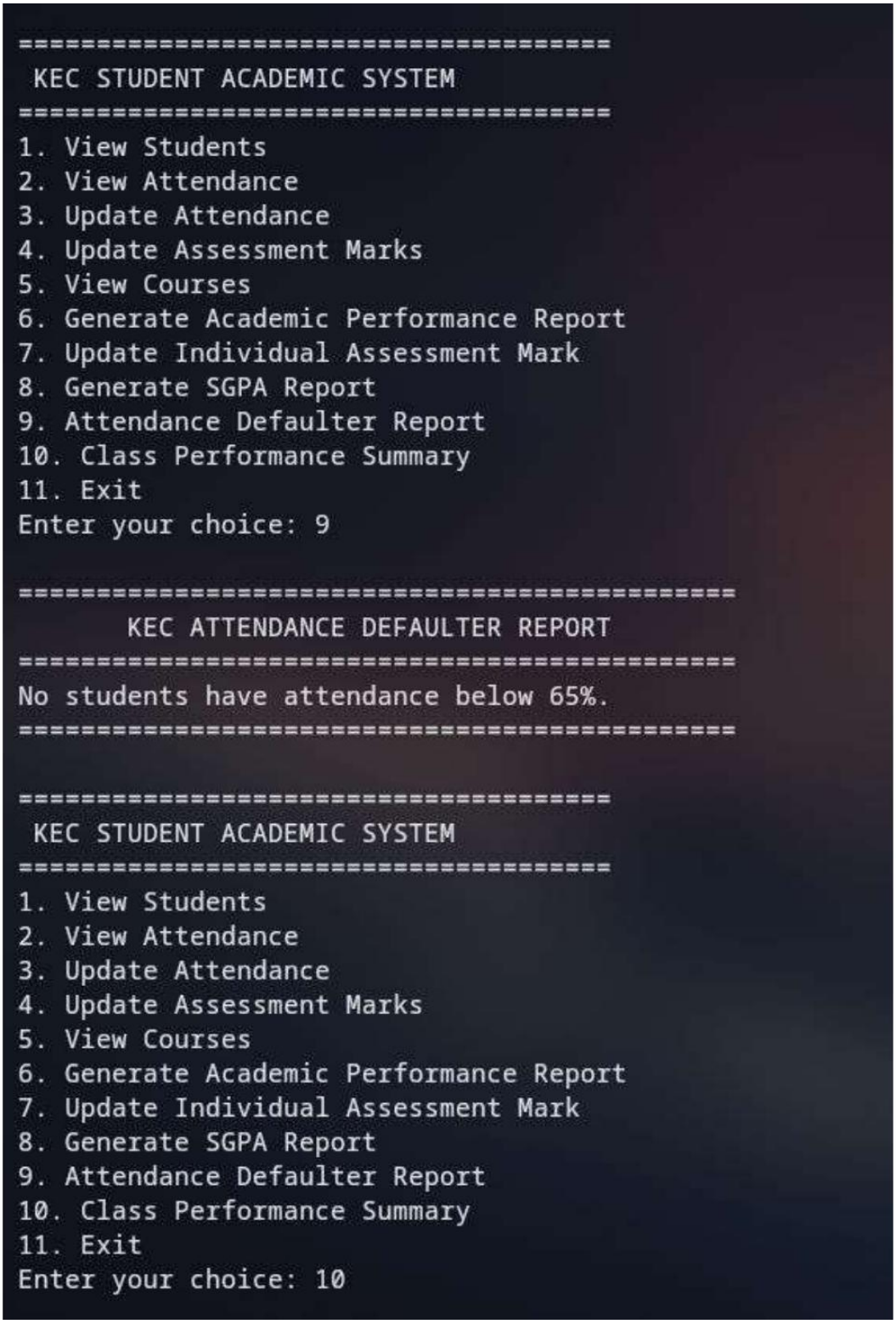
OUTPUT 4 – ACADEMIC PERFORMANCE REPORT



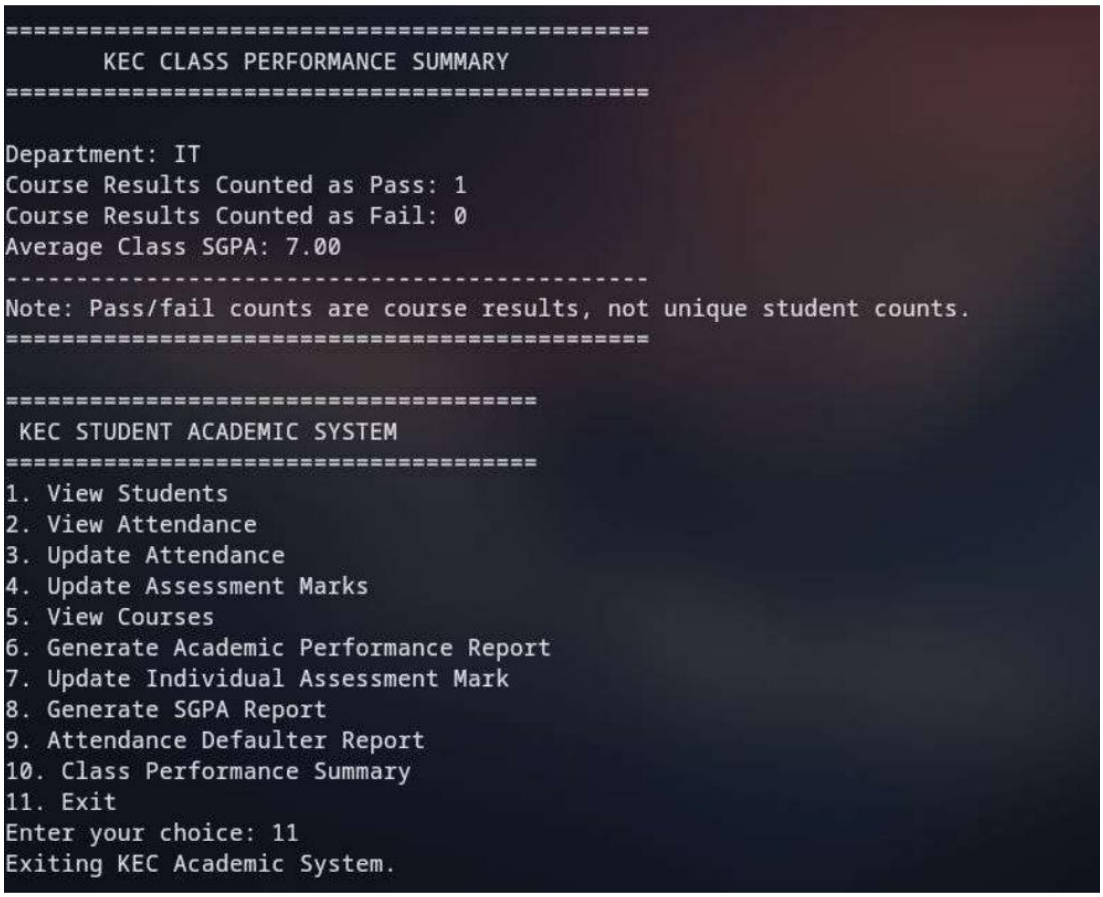
OUTPUT 5 – STUDENT PERFORMANCE AND SGPA REPORT



OUTPUT 6 – ATTENDANCE DEFAULTER REPORT



OUTPUT 7 – CLASS PERFORMANCE SUMMARY


==========
KEC CLASS PERFORMANCE SUMMARY
==========
Department: IT
Course Results Counted as Pass: 1
Course Results Counted as Fail: 0
Average Class SGPA: 7.00

Note: Pass/fail counts are course results, not unique student counts.
==========
==========
KEC STUDENT ACADEMIC SYSTEM
==========
1. View Students
2. View Attendance
3. Update Attendance
4. Update Assessment Marks
5. View Courses
6. Generate Academic Performance Report
7. Update Individual Assessment Mark
8. Generate SGPA Report
9. Attendance Defaulter Report
10. Class Performance Summary
11. Exit
Enter your choice: 11
Exiting KEC Academic System.

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MariaDB. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

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.https://www.youtube.com/results?search_query=Java+JDBC+MySQL+tutorial

10. Anthropic, Claude AI Assistant (used for project ideas and guidance).

https://claude.ai/

GITHUB:

LINK:	**https://github.com/yuvashreeb2007-hue/KEC-R2024-Student-Academic-Evaluation-System**